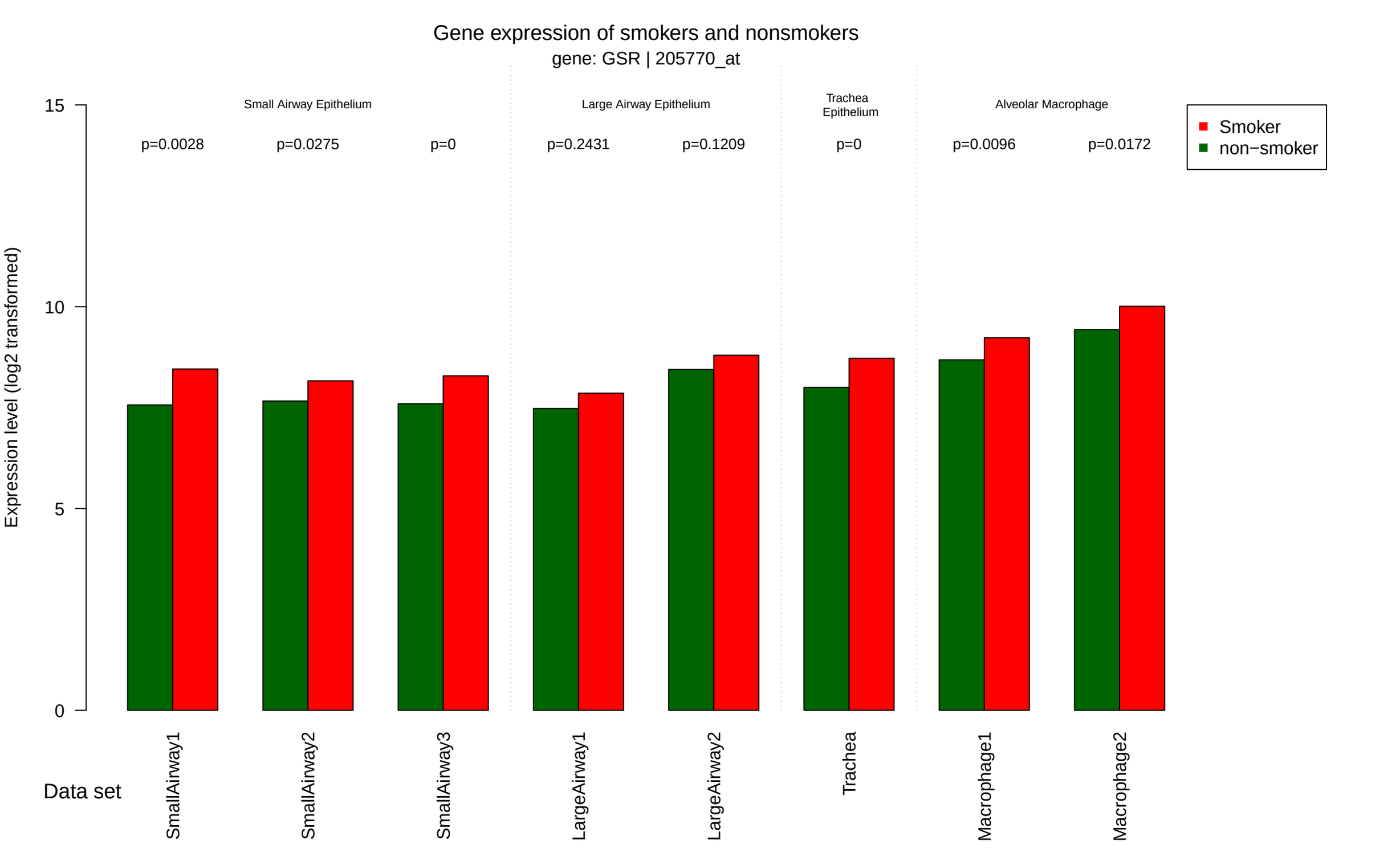
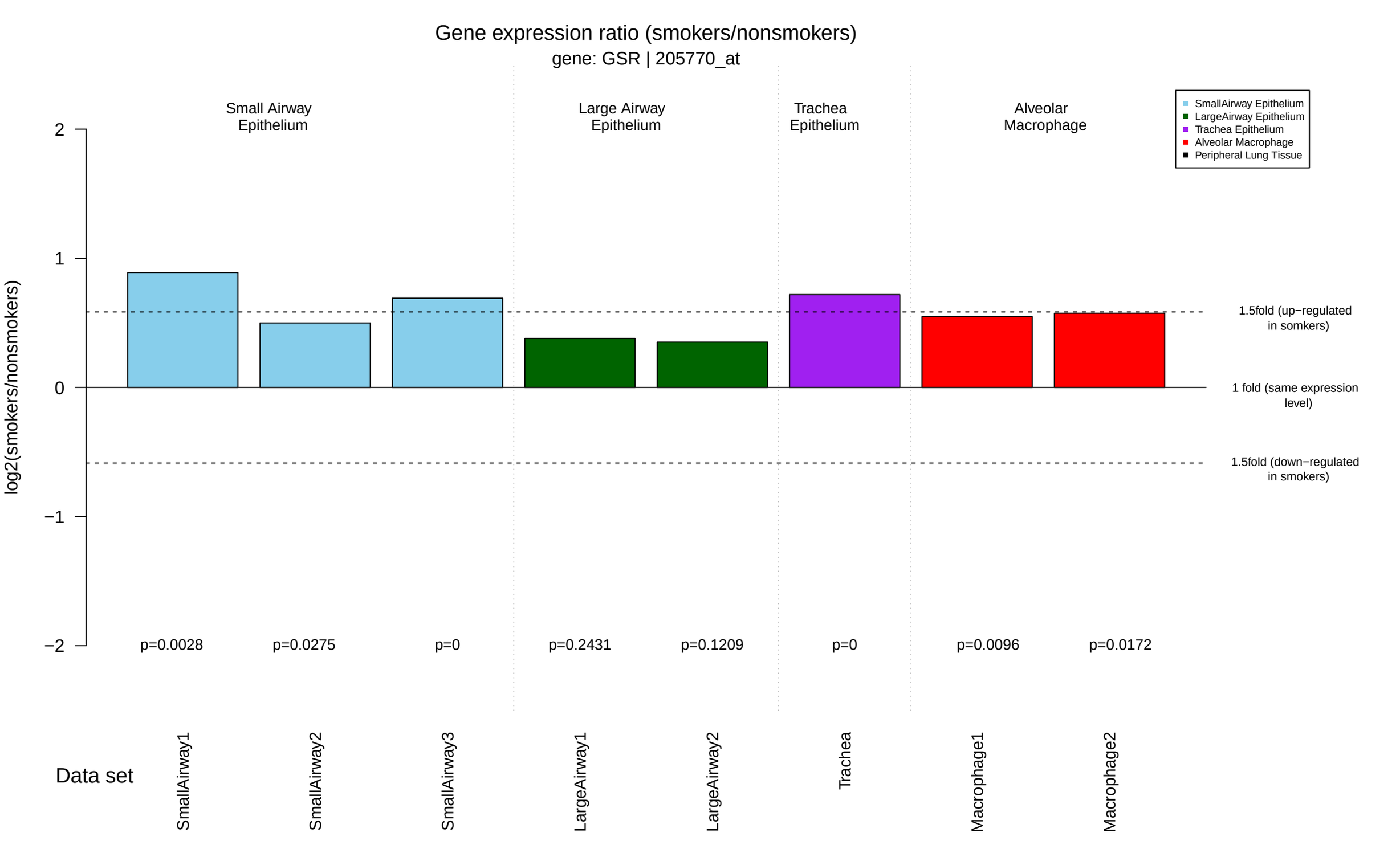
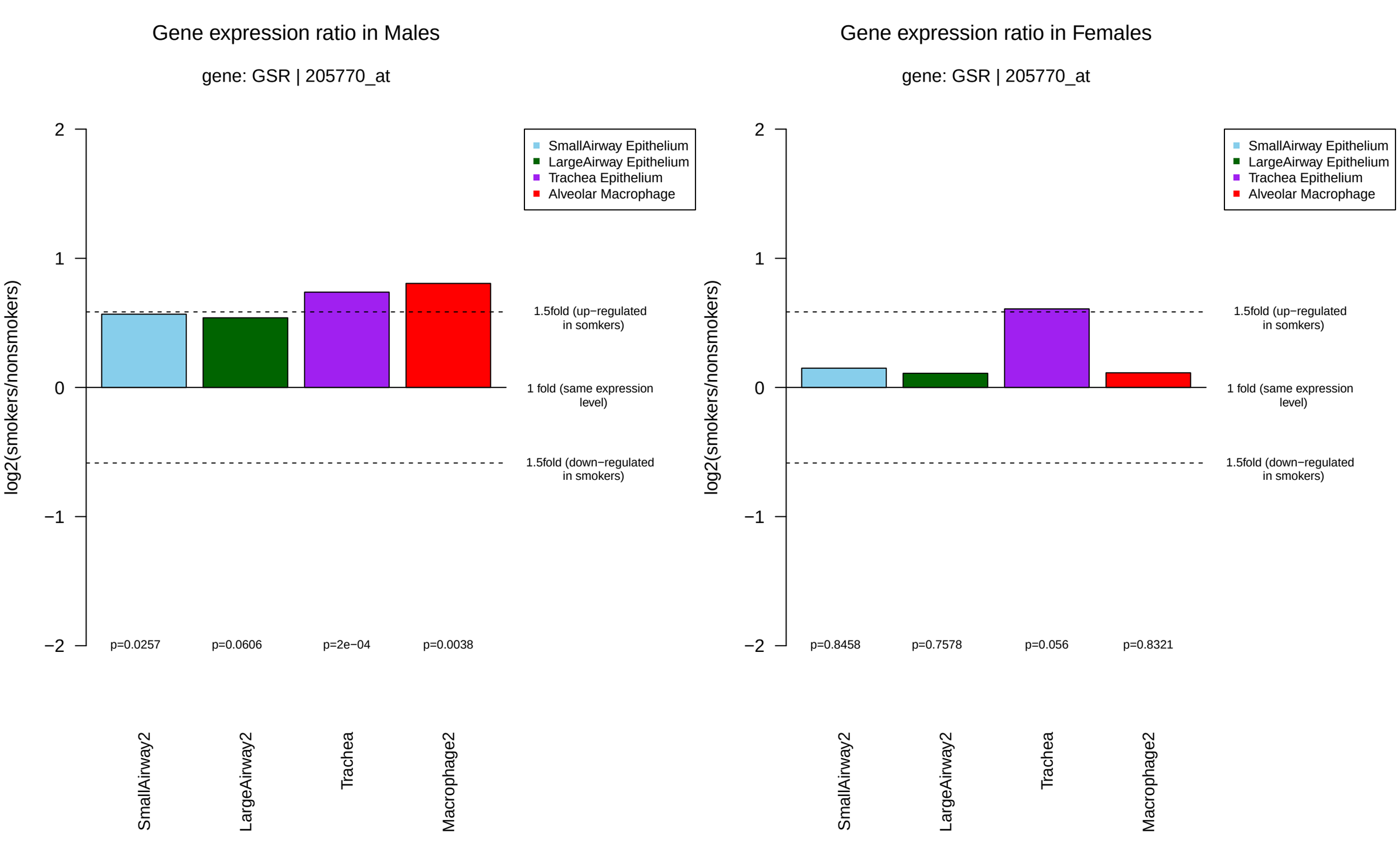
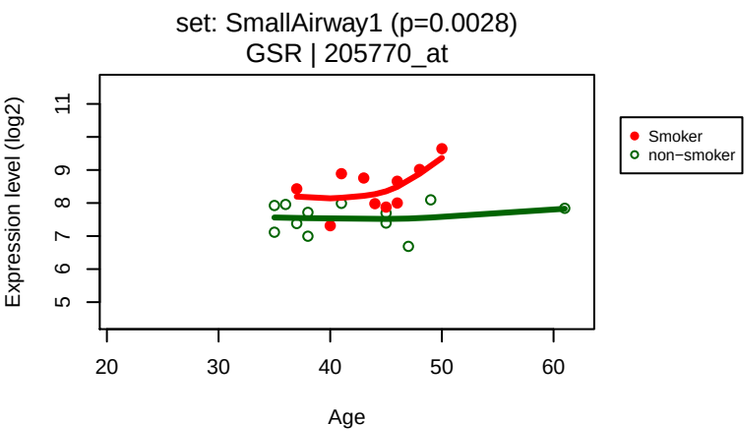
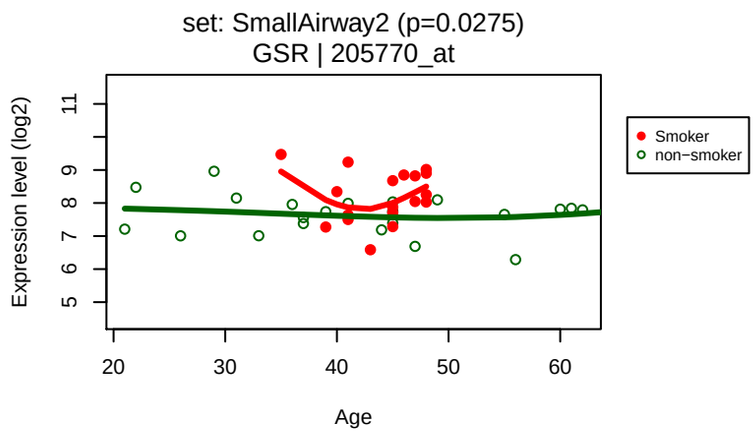
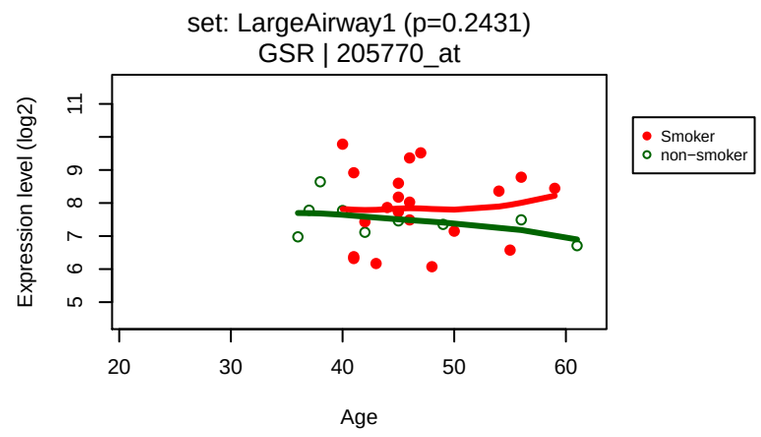

Gene expression of smokers and nonsmokers
gene: GSR | 205770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0028
p=0.0275
p=0
p=0.2431
p=0.1209
p=0
p=0.0096
p=0.0172
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

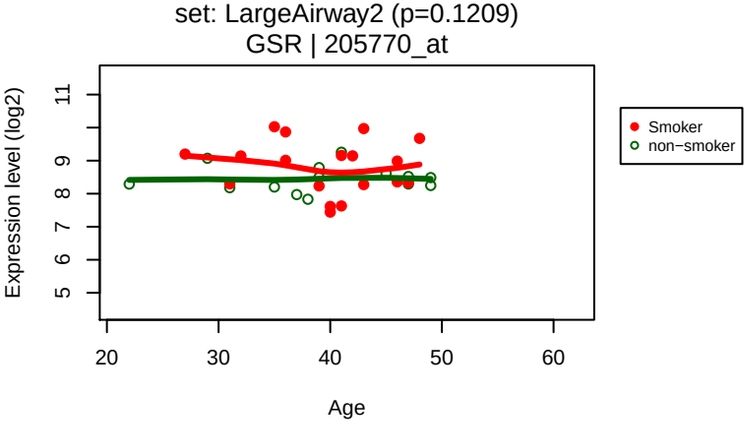

Gene expression ratio (smokers/nonsmokers)
gene: GSR | 205770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0028
p=0.0275
p=0
p=0.2431
p=0.1209
p=0
p=0.0096
p=0.0172
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

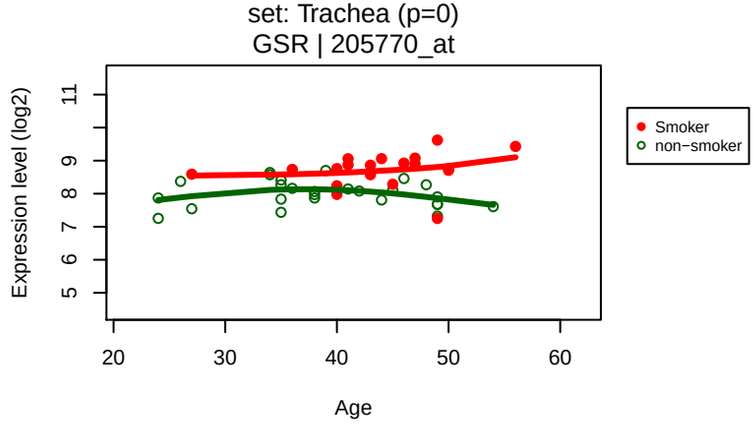

Gene expression ratio in Males
gene: GSR | 205770_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0257
p=0.0606
p=2e−04
p=0.0038
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSR | 205770_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8458
p=0.7578
p=0.056
p=0.8321
SmallAirway2
LargeAirway2
Trachea
Macrophage2

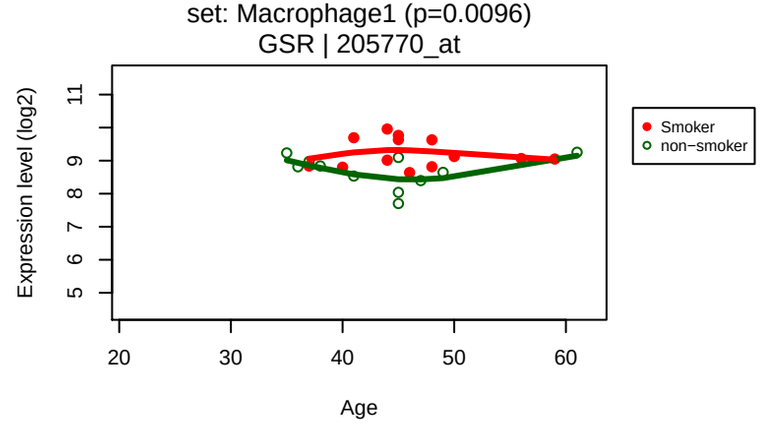

set: SmallAirway1 (p=0.0028)
GSR | 205770_at
Expression level (log2)
Age
Smoker
non−smoker

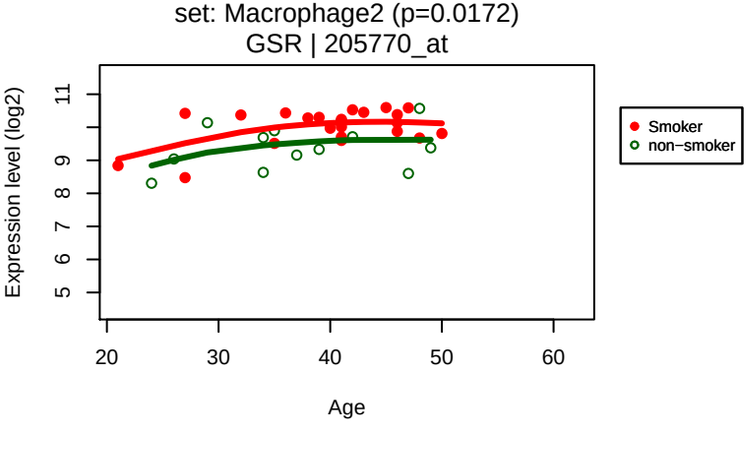

set: SmallAirway2 (p=0.0275)
GSR | 205770_at
Expression level (log2)
Age
Smoker
non−smoker

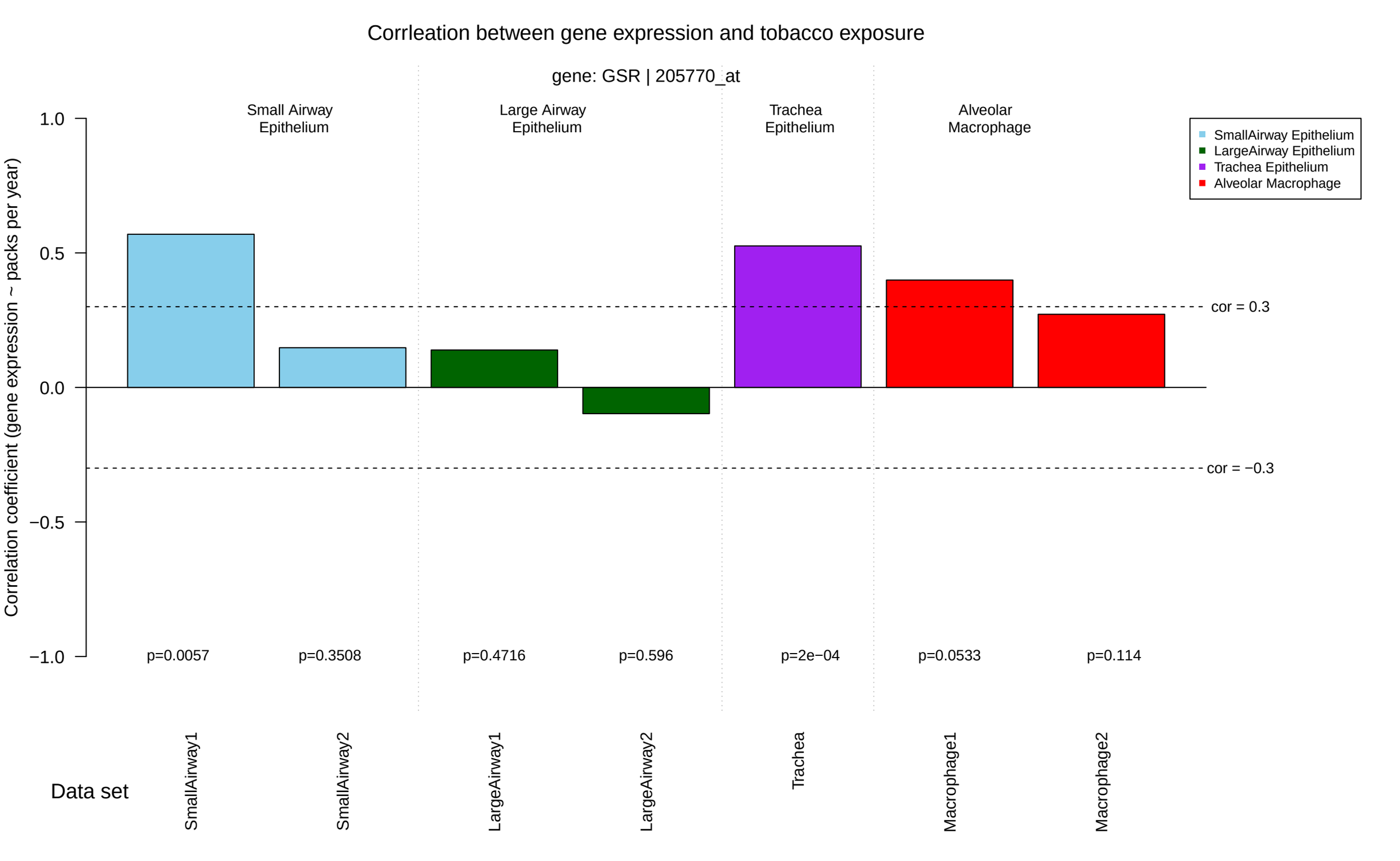

set: LargeAirway1 (p=0.2431)
GSR | 205770_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1209)
GSR | 205770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
GSR | 205770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0096)
GSR | 205770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0172)
GSR | 205770_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSR | 205770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0057
p=0.3508
p=0.4716
p=0.596
p=2e−04
p=0.0533
p=0.114
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2